

Diego Flores Martínez

Online Portfolio: <https://xxdiegofloresxx.github.io/MyPortfolio>

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EDUCATION

Monterrey, NL, México | **08/2021 - Present**

Instituto Tecnológico y de Estudios Superiores de Monterrey | Bachelor of Mechatronics Engineering

- Cumulative **GPA**: 3.60/4.00

Minor | Automotive Engineering

WORK EXPERIENCE

Hydraulics and Routing Design Intern | John Deere

Monterrey, NL, México | **09/2023 – 09/2025**

- Designed precise **3D models of hydraulic** systems for agricultural machinery utilizing **Creo Parametric**; ensured all technical drawings met industry standards to improve accuracy in the manufacturing workflow.
- Developed detailed flexible component simulations for specialized agricultural machinery using **Industrial Path Solutions**, identifying potential failure points and assembly issues in advance, which improved product reliability and reduced warranty claims.
- Participated in **Continuous Improvement** initiatives addressing recurrent structural flaws identified during testing, enhancing product **manufacturability** and **reducing** potential **warranty claims**.
- Participated in **EPDP** team initiatives focused on **product development**, contributing to the design and implementation of **hydraulic** and **pneumatic** systems for **seed** and **fertilizer delivery** applications.

SKILLS

Professional & Interpersonal Skills

- Project Management & Teamwork
- Collaboration & Communication
- Problem-Solving & Critical Thinking
- Time Management & Deadline-Driven
- Adaptability & Continuous Learning

Quality, Manufacturability & Process Improvement

- Root Cause Analysis (RCA)
- Lean Manufacturing & Six Sigma (basic)
- Design for Manufacturability (DFM) & Design for Assembly (DFA)
- Cost Reduction & Warranty Claim Mitigation
- Testing & Validation

Technical & Engineering Skills

- Advanced Mechanical & Structural Analysis (FEM: ANSYS, IPS)
- CAD: SolidWorks, Creo Parametric, CATIA
- Geometric Dimensioning & Tolerancing (GD&T)
- Material Science & Metallurgy
- Fixture & Tooling Design
- Electric & ICE Powertrain Knowledge, Power Electronics
- Battery Systems & Powertrain Integration (basic understanding)
- Electronics Systems Design & Analysis
- Automation, PLCs & Robotic Systems

Software, Simulation & Programming Tools

- Matlab & Simulink
- LTSpice, NI Multisim, KiCAD
- Automation Studio, FluidSIM
- CarSim
- Tecnomatix Plant Simulation & Process Simulate
- Programming / Data Analysis: Python, C++, LabVIEW

PROJECTS

Design Team Leader – F.I.R.S.T. Robotics Team.

Monterrey, NL, México | 08/2018 – 06/2021

FRC Design and Manufacturing Team Leader.

- Designed precise 3D models of robotic subsystems using CAD software, ensuring compliance with competition standards and streamlining the assembly process.
- Led a sub-team dedicated to the development of a specialized intake mechanism, managing both design and manufacturing phases to meet strict performance requirements.
- Implemented iterative design processes in collaboration with cross-functional teams, securing seamless integration of the intake system with overall robot architecture.
- Conducted subsystem testing and optimization, applying performance data and feedback to enhance mechanical reliability and competitive performance.

Junior Consultant – ITESM Industry Project with Parker.

Monterrey, NL, México | 03/2024 – 06/2024

New automatic assembly cell design.

- Achieved enhanced process reliability through the design of a fully integrated automatic assembly cell featuring safety measures and real-time monitoring capabilities using **Cognex technology**.
- Developed custom **feeders and fixtures** for automated presses, which reduced manual intervention by 60% and decreased production cycle time, enhancing overall operational efficiency and consistency.

Junior Consultant – ITESM Project with Piaggio.

Monterrey, NL, México | 08/2024 – 12/2024

Combustion Engine Powertrain Analysis and Electric Engine Retrofitting Proposal

- Analyzed combustion and electric powertrains using the SuperFlow dynamometer, achieving retrofit strategies within $\pm 5\%$ power variance.
- Designed geometric layouts for three-wheeled vehicle powertrains, improving integration efficiency by 20% while maintaining modularity.
- Reduced battery train weight by 10%, leading to lower energy consumption and improved performance. Performed statistical analysis in Excel to optimize rolling resistance and aerodynamic coefficients.

Junior Consultant – ITESM Project with Ternium & Metalsa.

Monterrey, NL, México | 08/2024 – 12/2024

Modular chassis design for ICE, EV and hybrid electric vehicles.

- Utilized **ANSYS Workbench** to analyze chassis geometry, uncovering design weaknesses and enhancing the structure's modularity to support both internal combustion and electric vehicle platforms.
- Leveraged advanced **static physics principles** within **SOLIDWORKS** to refine chassis geometry, realizing a **15% weight reduction** and improving modularity for greater assembly efficiency and design versatility.
- Reviewed Ternium's iron material options, selecting the most suitable material based on mechanical properties; the chosen material provided a 15% weight reduction without compromising structural strength.

ACHIEVEMENTS

- Participated in national, regional and global robotics competitions as part of a high-performing team, contributing to four separate championship victories through strategic design and manufacturing leadership (2018-2021).
- **Received recognition** from **John Deere** for developing an innovative fixture proposal to bend a critical plate, significantly improving manufacturing precision and workflow (2023).
- **Received acknowledgment** from **Parker** for a groundbreaking manufacturing cell design, addressing key operational challenges while contributing findings that targeted the three biggest causes of delays in the assembly process (2024).
- **Received recognition** from **ABB** for the design of a robotic welding cell, addressing key operational challenges and enhancing production efficiency (2024).
- Accomplished the first **internship at John Deere**, contributing to advanced projects in agricultural machinery that streamlined development processes and enhanced product efficiency for future innovations (2025).